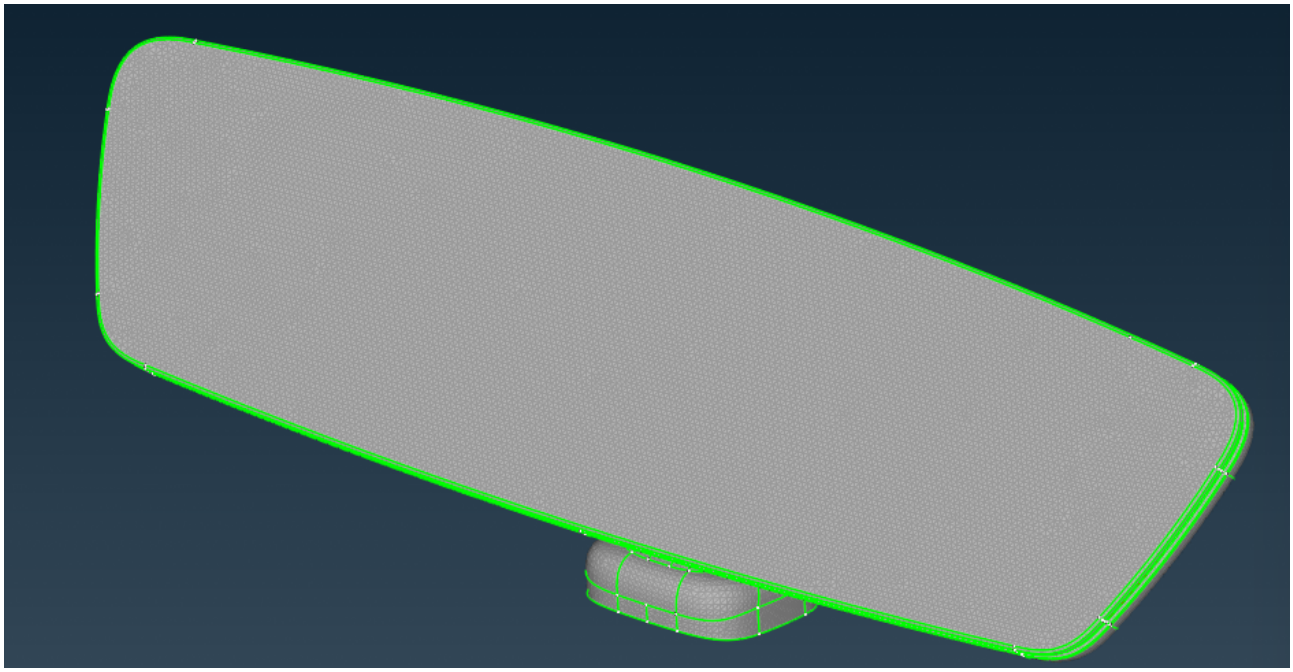
Rear-View Mirror

Surface and Tetrahedral Volume Meshing in ANSA

Mini Project Report | BETA CAE ANSA v25.0.0 | Unstructured mesh with Tetra Rapid



Shell elements (TRIA3)	Solid elements (TETRA4)	Target length	Tet collapse limit
72,512	388,468	1 mm	0.2

1. Objective

The aim was to take the supplied IGES model of a car rear-view mirror (housing, stem and ball-joint mount), check it for geometrical errors, mesh its surfaces to a given 2D quality standard, and fill the closed volumes with a tetrahedral mesh that passes the 3D quality check without kinks or dips on the surface.

2. Quality criteria

Criterion	Value	Applies to
Target / average length	1 mm	2D
Minimum length	0.5 mm	2D
Maximum length	2 mm	2D
Tet collapse	0.2	3D

Only min/max length is checked on 2D elements and only tet collapse on 3D elements, as stated in the problem.

3. Workflow

- **Import** the IGES geometry into ANSA.
- **Geometry check and cleanup:** topology, free edges, small faces, and merging faces into larger macros.
- **Mesh parameters:** target 1 mm, min 0.5 mm, max 2 mm, tria elements.
- **2D meshing:** tria surface mesh with local length control on grooves, fillets and corners.
- **Quality fixing:** failed elements (min/max length) remeshed locally.
- **Volume definition:** closed shell set defined as volumes.
- **3D meshing:** Tetra Rapid on all volumes.
- **Final checks** on tet collapse and surface smoothness, then save as .ansa.

4. Geometry check and cleanup

The mesh is only as good as the geometry under it, so the model was inspected first. Free edges (shown in red) and small faces split the surface into many macros, which would force tiny elements. These were cleaned so the housing, grooves, stem and ball joint form a closed, consistent topology ready for meshing.

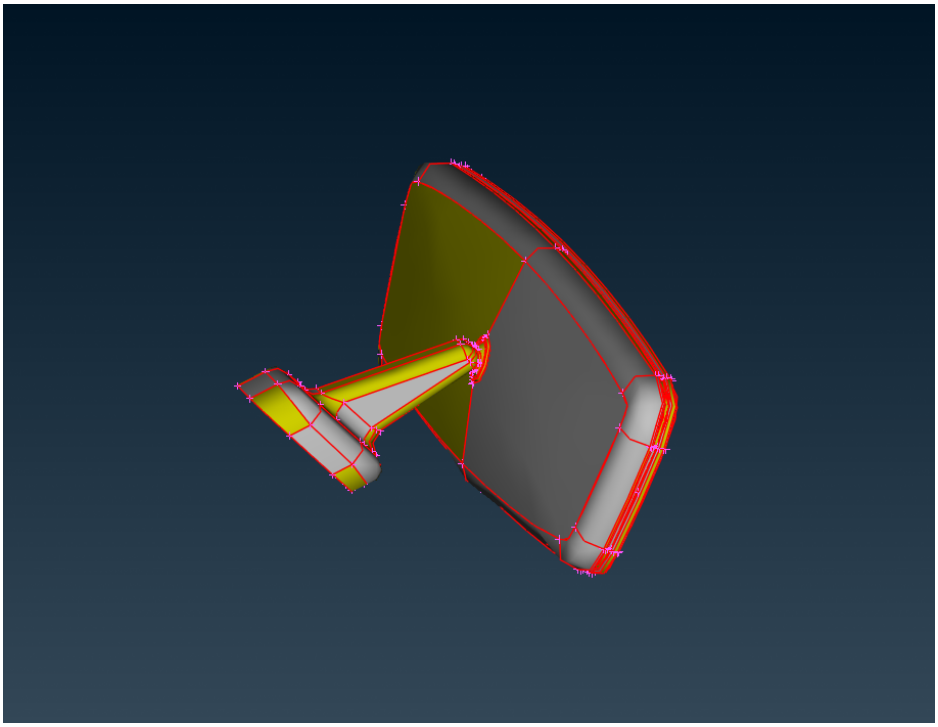


Figure 1. Imported geometry with face boundaries and free edges (red) shown.

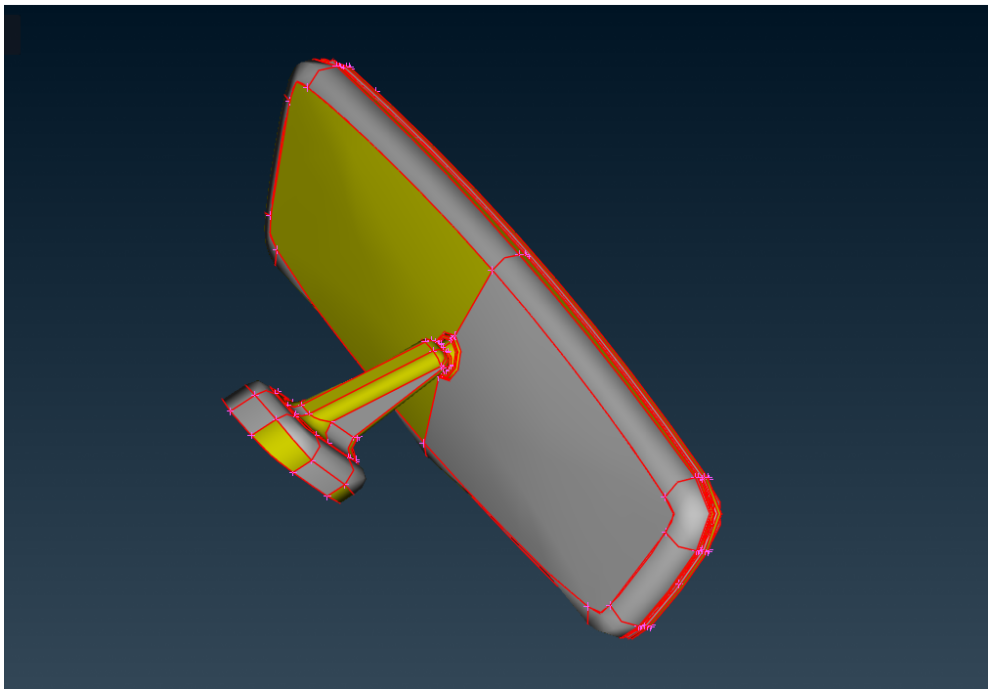


Figure 2. Second view of the same geometry, showing the stem and mount joining the housing.

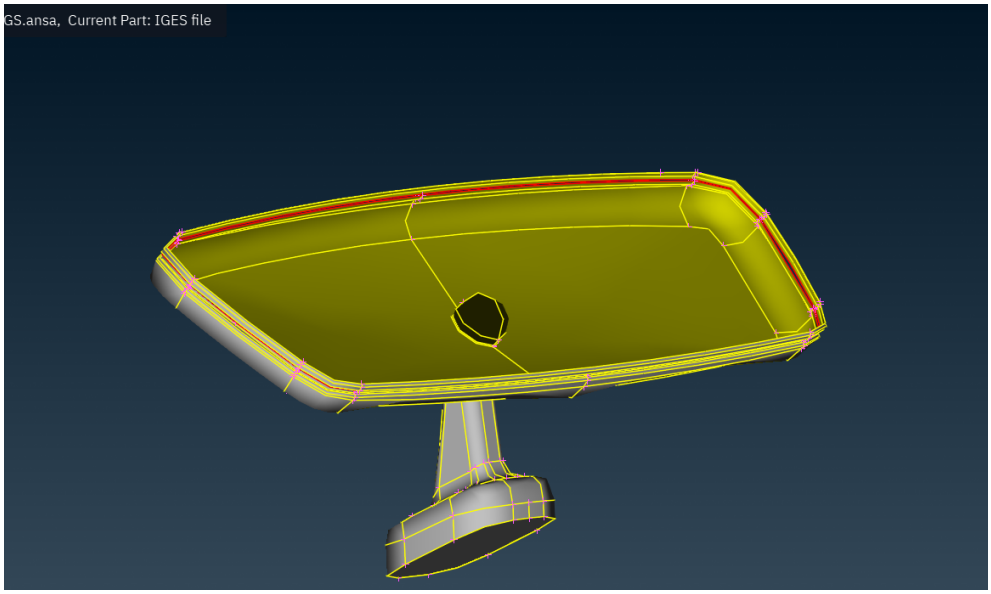


Figure 3. Back side of the mirror housing during cleanup; the highlighted edge (red) is an entity under inspection.

Mid-surface note

The component is a solid body, not a thin sheet. A mid-surface extraction is not needed here: the outer skin is meshed with trias and the interior is filled with tetras. Mid-surface extraction is appropriate for thin-walled parts that would be meshed with shells only.

5. 2D meshing approach

The surface was meshed with tria elements at the 1 mm target length. The aim was a uniform, smooth surface mesh because every error in it carries into the tet mesh below.

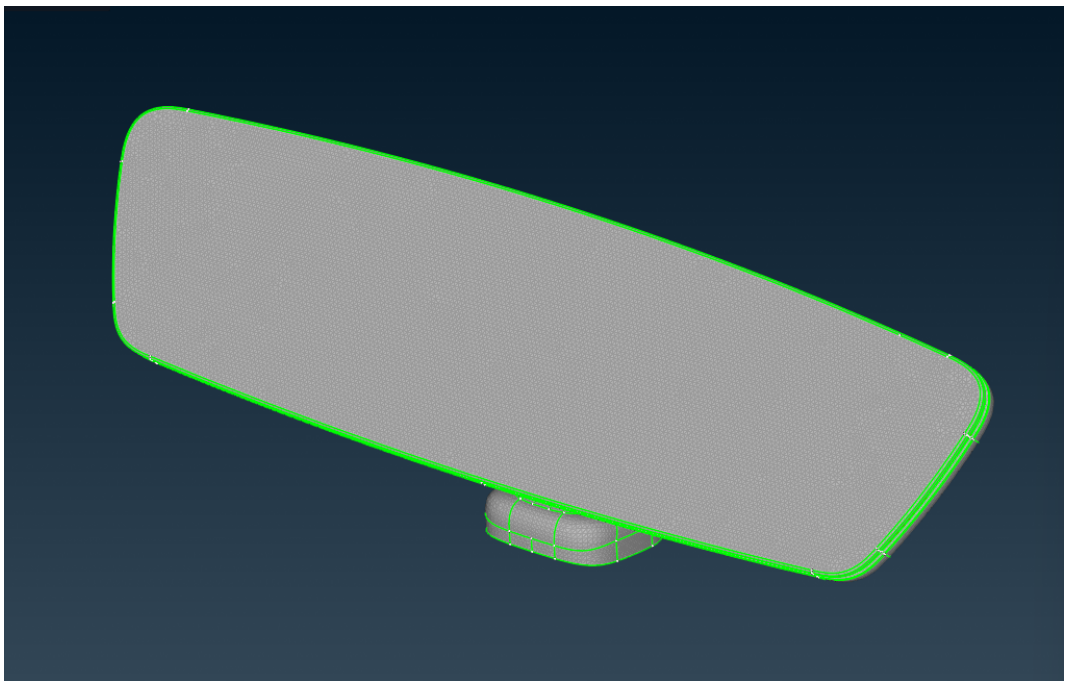


Figure 4. Tria surface mesh of the full mirror face.

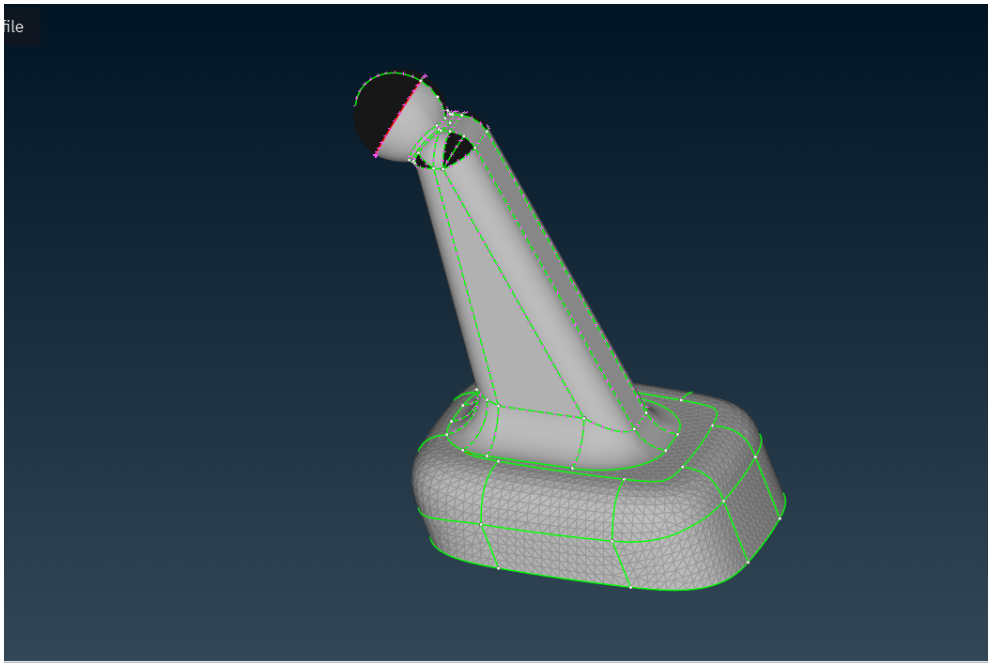


Figure 5. Tria mesh on the stem and ball-joint mount.

Corners and fillets compared with flat faces

Different mesh treatment was used by region. On the **corners and fillet surfaces**, where the surface curves quickly, ortho-tria meshing was used so the trias stay regular and follow the curvature, which keeps the corner profile accurate and avoids stretched elements. On the **flat faces**, regular tria meshing is enough because the surface does not bend, so there is no curvature to capture.

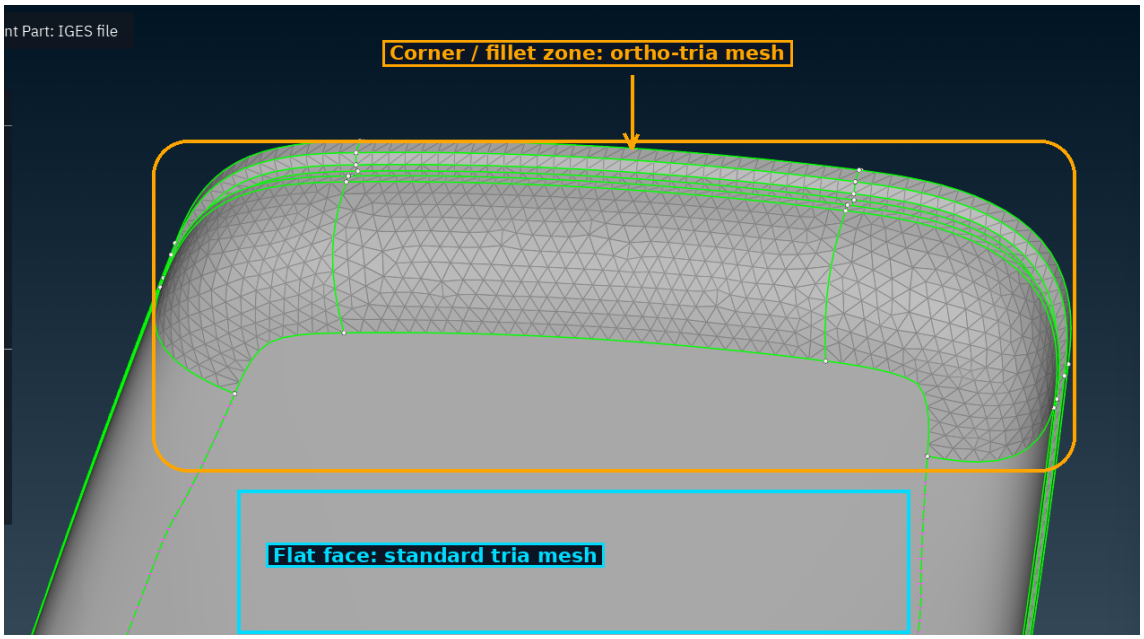


Figure 6. Orange: corner and fillet zone with ortho-tria mesh. Blue: flat face with standard tria mesh.

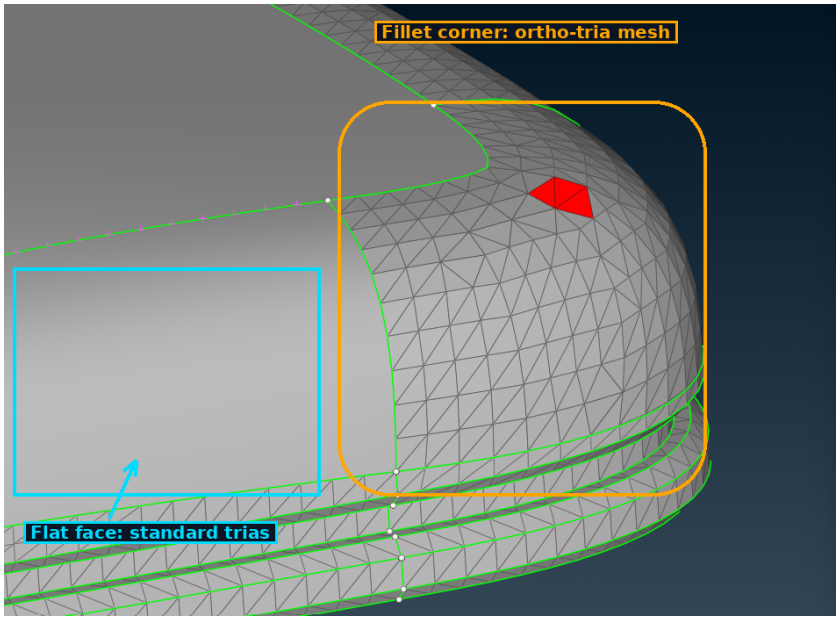


Figure 7. Corner detail. The red element marks a failed length criterion that was found and fixed by local remeshing.

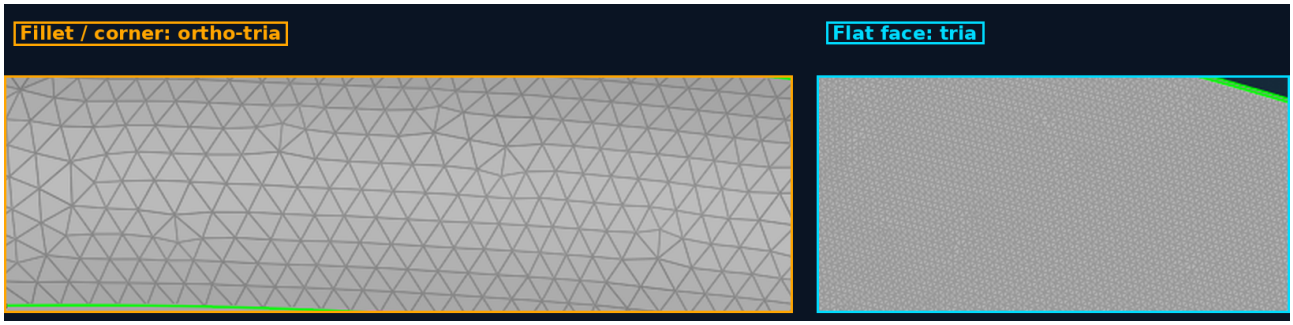


Figure 8. Side-by-side close-up of the two regions. Note: the two crops are at different zoom levels, so absolute element size is not comparable here.

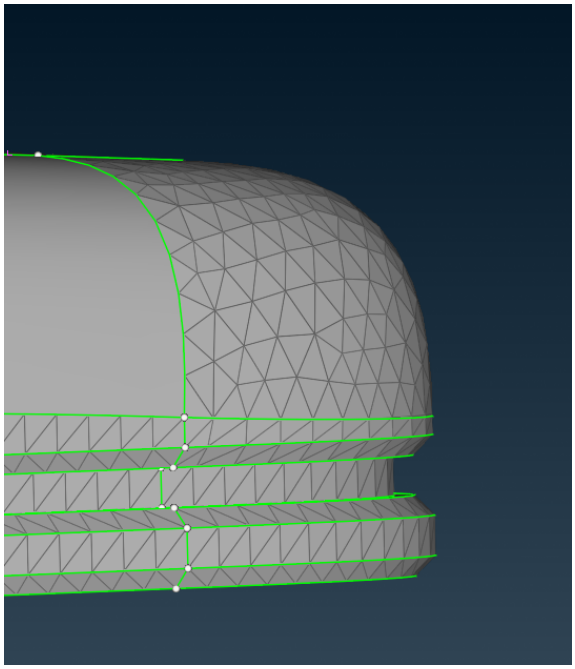


Figure 9. Side groove on the housing edge. The mesh follows the small grooves, which needed local length control.

6. Volume meshing with Tetra Rapid

Unstructured mesh. Elements have no regular grid pattern. This suits curved, organic shapes such as the mirror housing, grooves and ball-joint stem. A structured hexahedral mesh would need heavy geometry decomposition to fit them.

Tetra Rapid. ANSA's fast tetrahedral mesher builds the volume mesh directly from a closed, good-quality shell mesh. It is quick on large models and gives good control over element size. Tetra FEM is an alternative that can give higher element quality at the cost of speed; Tetra Rapid was chosen here because the surface mesh was already controlled and the model is large (about 388,000 solids).

Prerequisite. The 2D mesh must be closed, with no free edges, overlaps or intersections. That is why cleanup and 2D quality came first.

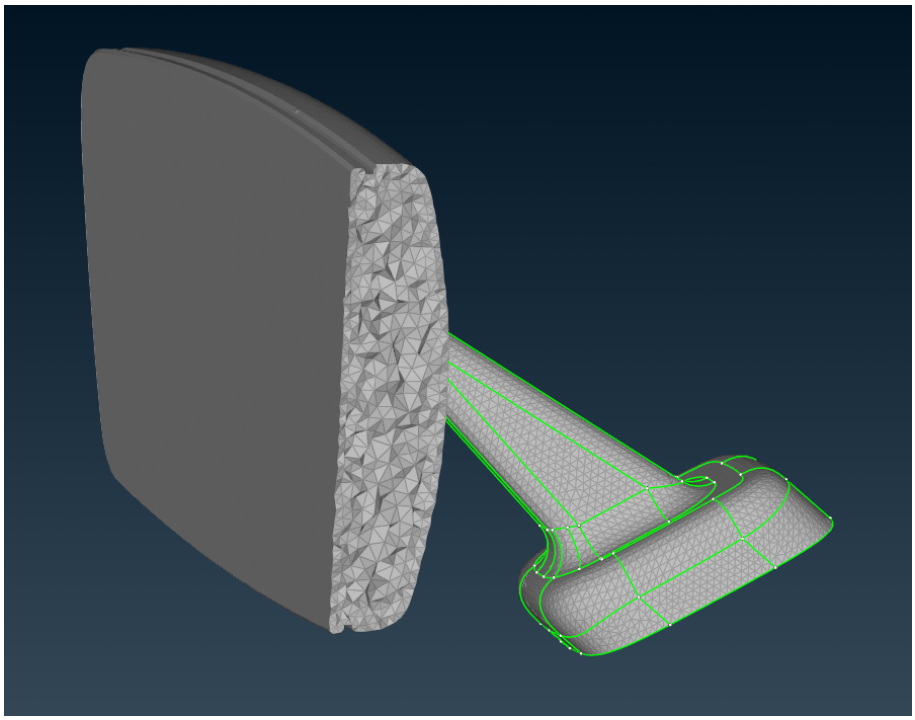


Figure 10. Cut-section of the housing: tetrahedral elements fill the interior solid.

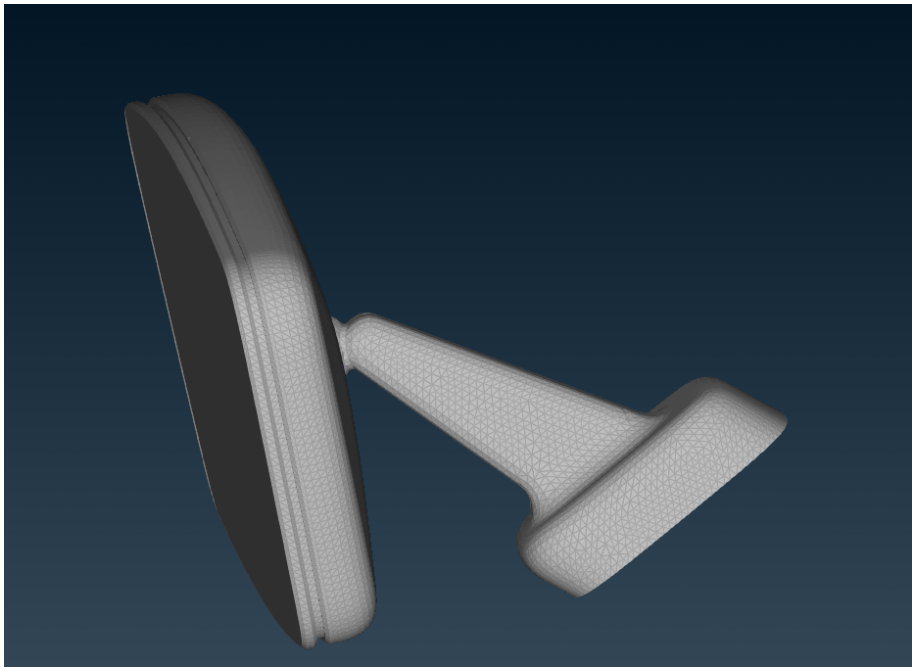


Figure 11. Side view of the volume-meshed mirror, stem and mount.

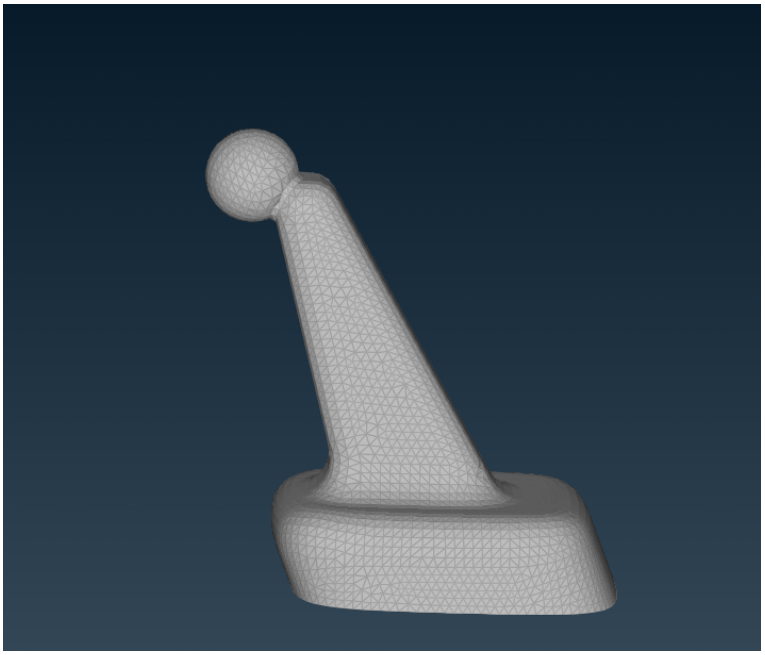


Figure 12. Ball-joint mount mesh, smooth with no visible kinks or dips.

7. Results

Metric	Value
Shell elements (TRIA3)	72,512
Quad elements	0
Solid elements (TETRA4)	388,468
Target length	1 mm
2D checks	Min 0.5 mm, Max 2 mm
3D check	Tet collapse 0.2

Shells	72512
Quads	0
Trias	72512
Solids	388468
Tetras	388468

Figure 13. ANSA Quality Monitor for the final model.

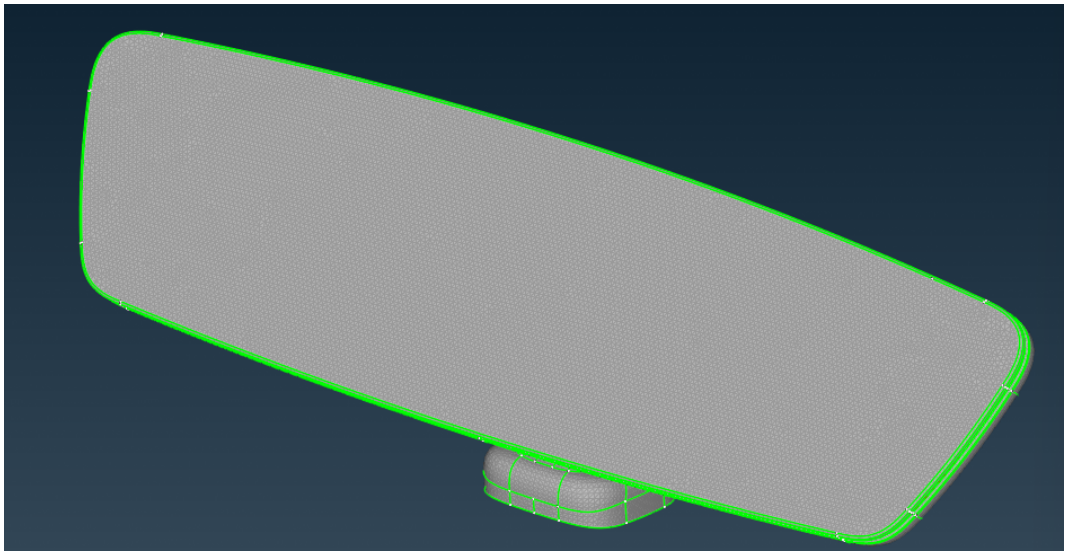


Figure 14. Final meshed mirror.

8. Challenges and fixes

- **Grooves on the housing edge:** needed local length control so the mesh follows the narrow profile.
- **Failed elements at corners:** min/max length violations were fixed by remeshing the affected macros.
- **Unmeshed macros and unchecked entities:** cleared before volume meshing.
- **Surface kinks after volume meshing:** avoided by achieving good 2D quality before running Tetra Rapid.

9. What I learned

- Clean geometry decides whether meshing succeeds.
- Mesh length is controlled through perimeters and mesh parameters.
- Curved corners and fillets need a different mesh treatment from flat faces.
- 2D quality drives 3D quality: good trias give good tets.
- How to define volumes from closed shells and run Tetra Rapid.
- How to read the Quality Monitor and fix length and tet collapse violations.
- How to debug unmeshed macros, off elements and unchecked entities.

10. Tools

BETA CAE ANSA v25.0.0, IGES geometry, Tetra Rapid, NASTRAN deck.